

FACULTY OF INFORMATION TECHNOLOGY

2026 MOBILE APP DEVELOPMENT 700

ASSIGNMENT

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QUESTION	EXAMINER MARKS	MODERATOR MARKS	REMARKS

ASSIGNMENT INSTRUCTIONS

Please tick each box to confirm completion. (10 Marks)

Use Times New Roman font, size 12, with 1.5 line spacing throughout the document. Apply
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For essay style assignments, please include the following sections:

Table of Contents Introduction

Main Body (with relevant subheadings) Conclusion

References

Submit the assignment in PDF format on Moodle. Use
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Include a signed declaration of originality.

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I hereby declare that this assignment is my own work and has not been copied from any other source except where due acknowledgment is made. I affirm that all sources used have been properly cited and that this submission complies with the institution's policies on academic integrity and plagiarism.

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Introduction

Food waste remains a significant challenge in many households due to poor inventory management and a lack of awareness regarding available ingredients. Consumers frequently purchase duplicate products, allow ingredients to expire, or fail to utilize food items efficiently. Mobile technology presents an opportunity to address these challenges through applications that assist users in tracking available ingredients and generating meal suggestions based on pantry contents.

The Smart Pantry Manager application was developed to assist users in managing pantry ingredients and discovering recipes that can be prepared using ingredients already available in their homes. The application allows users to add, edit, view and remove ingredients from a pantry inventory while storing all information in a local database. Based on the ingredients available, the application evaluates recipes and displays only those recipes that can be prepared using the user's current pantry contents.

The application was developed using Java in Android Studio and uses the Room persistence library to provide permanent data storage. The solution demonstrates key mobile development concepts including activities, RecyclerViews, Room Database integration, input validation, navigation, data persistence and application lifecycle management. Room Database simplifies local database implementation in Android applications (Android Developers, 2026).

System Overview

The Smart Pantry Manager application consists of three primary modules:

1. The first module is Pantry Management. This module allows users to create, update, view and delete pantry items. The pantry inventory serves as the primary data source used throughout the application.

2. The second module is Recipe Management. Recipe information is stored within the application and linked to required ingredients. The system evaluates recipe requirements against the pantry inventory to determine whether recipes can be prepared.
3. The third module is Settings Management. This module provides utility functions such as clearing all pantry ingredients and displaying application information.

The application operates entirely offline and stores all information locally through an SQLite database implemented using Room Database.

Application Requirements

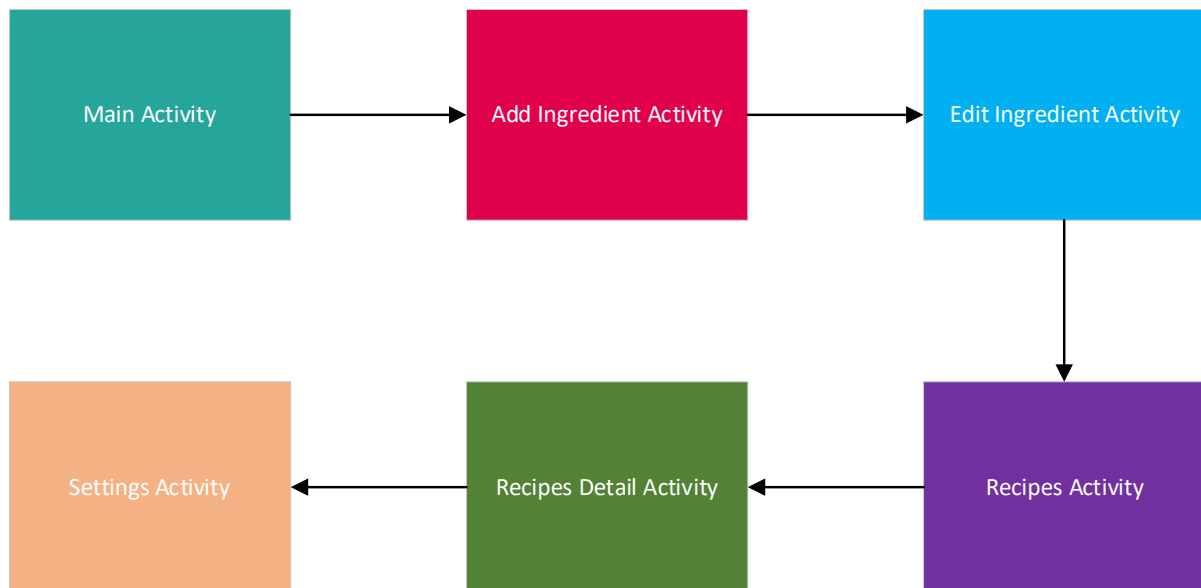
The following requirements were identified for the project:

- The application must allow users to add pantry ingredients.
- The application must support viewing stored ingredients through a RecyclerView.
- The application must support updating ingredient information.
- The application must support deleting ingredients.
- The application must permanently store ingredient information using a database.
- The application must display recipes based on available ingredients.
- The application must prevent recipes from appearing if required ingredients are unavailable.
- The application must include multiple screens connected through navigation.
- The application must contain a settings interface.
- The application must implement user input validation.

System Design

The application was designed using a multi-activity architecture. Material Design components assist developers in creating consistent and user-friendly interfaces (Android Developers, 2023).

Application Flow



The user enters ingredient data through dedicated input screens. Data is stored within Room Database and retrieved through Data Access Objects (DAO). Recipes are evaluated against stored ingredient records and displayed to the user when requirements are satisfied.

Database Design

The application uses Room Database to provide permanent storage for ingredients and recipes.

Three database entities were implemented.

Ingredient Table

Ingredient
Id
Name
Quantity

unit
expiryDate

Recipe Table

Recipe
Id
recipeName
instructions

Recipe Ingredient Table

RecipeIngredient
Id
recipeId
ingredientName

The RecipeIngredient table forms the relationship between recipes and required ingredients.

Application Implementation

The application was developed in Java using Android Studio. Several Android development concepts were implemented:

Activities

The application contains multiple activities responsible for specific functions. Activities serve as the primary entry point for user interaction within Android applications (Android Developers, 2026).

Examples include:

- MainActivity
- AddIngredientActivity
- EditIngredientActivity
- RecipesActivity
- RecipeDetailActivity
- SettingsActivity

RecyclerView

RecyclerView was used to display:

- Pantry ingredients
- Available recipes

RecyclerView improves performance by recycling views and efficiently rendering large datasets. RecyclerView provides an efficient mechanism for displaying large datasets using reusable views (Android Developers, 2026).

Room Database

Room Database was selected because it simplifies SQLite integration and provides an object-oriented approach to data storage.

The implementation includes:

- Entities
- DAOs
- AppDatabase singleton

- CRUD functionality

Pantry Management Features

The pantry management module provides complete CRUD functionality.

Create Ingredient

Users can enter:

- Ingredient Name
- Quantity
- Unit
- Expiry Date

Information is validated before being saved to the database.

View Ingredients

Saved ingredients are displayed through a RecyclerView.

Each ingredient displays:

- Ingredient Name
- Quantity
- Unit
- Expiry Date

Update Ingredient

Users can select an existing ingredient and modify stored values.

Changes are immediately reflected in the RecyclerView.

Delete Ingredient

Users can remove ingredients through a confirmation dialog.

Deleted records are permanently removed from the Room Database.

Recipe Recommendation Features

The Recipe module represents the core functionality of the Smart Pantry Manager.

Recipes are seeded into the database during application initialization.

Examples include:

- Cheese Omelette
- French Toast
- Cheese Sandwich
- Pancakes
- Scrambled Eggs
- Egg Fried Rice
- Rice Bowl
- Tomato Soup
- Fruit Smoothie
- Chicken Curry
- Vegetable Stir Fry

Each recipe has associated ingredients stored within the RecipeIngredient table.

Recipe Matching Algorithm

The application determines recipe availability by comparing required recipe ingredients with pantry inventory items.

Pseudo-code:

- FOR each recipe - retrieve recipe ingredients
- FOR each recipe ingredient - check pantry
- IF ingredient missing - recipe unavailable
- IF all ingredients available - display recipe

Only recipes with complete ingredient matches are displayed to the user.

Recipe Detail Screen

The Recipe Detail screen displays:

- Recipe Name
- Required Ingredients
- Preparation Instructions

Settings and Additional Features

The Settings module provides utility features.

Clear Pantry

Users can remove all stored ingredients using a confirmation dialog.

About Application

The application provides version information and a brief project description.

Validation

The application validates input fields to prevent:

- Empty ingredient names
- Missing quantities
- Invalid records

Error messages are displayed where necessary.

Testing and Validation

Functional testing was performed across all modules.

Feature	Test Result
Add Ingredient	Passed
View Ingredients	Passed
Edit Ingredient	Passed
Delete Ingredient	Passed
Database Persistence	Passed
Recipe Matching	Passed
Recipe Details	Passed
Settings Screen	Passed
Clear Pantry	Passed
About Application	Passed

Feature	Test Result
Input Validation	Passed

Database persistence was verified by closing and reopening the application and confirming that stored ingredient information remained available.

Challenges Encountered and Solutions

One significant challenge involved configuring Room Database and establishing relationships between recipes and ingredients. This was resolved through the implementation of dedicated entity classes and foreign key references using RecipeIngredient records.

Another challenge involved creating the recipe matching algorithm. The algorithm initially displayed unrelated recipes. This issue was resolved by implementing strict ingredient comparison logic that ensured every required ingredient existed before a recipe was displayed.

Device emulator configuration also created development challenges. Testing was ultimately completed using a physical Android device which provided a more reliable development environment.

Reflection and Learning Outcomes

This project provided extensive practical experience in Android application development. Intent objects allow communication and navigation between application components (Android Developers, 2026).

Several concepts were explored in depth, including:

- Android Activities

- RecyclerView Implementation
- Room Database
- CRUD Operations
- Input Validation
- Intents and Navigation
- User Interface Design

The project also improved problem-solving abilities through the identification and resolution of configuration issues, database challenges and application logic requirements.

Conclusion

The Smart Pantry Manager application successfully satisfies the project requirements by providing pantry management and recipe recommendation functionality. The application enables users to manage ingredients efficiently, reduce food waste and discover recipes based on available pantry items.

The implementation demonstrates effective use of Android Studio, Java, Room Database and RecyclerView components while maintaining a user-friendly interface. The project highlights the practical application of mobile development concepts and provides a scalable foundation for future enhancements such as cloud synchronization, barcode scanning and nutritional analysis.

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